

Editor-in-Chief
Computers & Security

Re: Submission of “NARCIS: Authenticated Neural Coverless Image Signaling with Attack-Qualified Codebooks and Error Correction”

Dear Editor,

Please consider the enclosed manuscript for publication as a Research Paper in *Computers & Security*.

The manuscript addresses an operational limitation of selection-based coverless communication. A stable image descriptor alone does not ensure that a finite image index can realise every protected message symbol, survive channel processing, and recover an authenticated plaintext. NARCIS integrates a self-supervised image descriptor, balanced quantile codebooks, multi-attack cover qualification, keyed symbol assignment, AES-GCM protection, replay rejection, and Reed–Solomon correction into one end-to-end protocol that transmits unchanged natural images.

Evaluation uses five deterministic partitions of BOSSBase and five of Caltech-101, covering 512×512 grayscale photographs and variable-resolution RGB images. All 1,575 message-condition trials recovered the authenticated plaintext. Mean symbol accuracy was 98.74% on BOSSBase and 98.59% on Caltech-101. A learned selection CNN produced mean AUCs of 0.515 and 0.530, with both 95% confidence intervals including chance performance. Removing attack-based cover qualification in a controlled ablation reduced complete-message success from 100% to 43.65%.

The manuscript does not claim the highest published nominal capacity or universal resistance to steganalysis. It positions the contribution as a security-aware coverless protocol with measured robust operating points, explicit finite-index feasibility, authenticated recovery, and reproducible failure criteria. We believe this focus is aligned with the journal’s scope in secure multimedia communication, active-channel robustness, and empirical security evaluation.

The manuscript is original, is not under consideration elsewhere, and has been approved by all authors. The authors declare no competing interests. The public project archive contains source code, deterministic seeds, consolidated results, plotting scripts, tests, and attack definitions. Datasets and trained checkpoints are not redistributed.

Thank you for considering this submission.

Sincerely,

Stéphane Gaël R. Ekodeck

Corresponding author

University of Yaoundé I, LIRIMA, Team GRIMCAPE

stephane-gael.ekodeck@facsciences-uy1.cm